

Ayush Urs

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EDUCATION

University of California, San Diego

San Diego, CA

M.S. Computer Science

University of California, San Diego

San Diego, CA

B.S. Mathematics – Computer Science

GPA: 3.89 September 2023 – March 2026

Relevant Coursework: Machine Learning: Algorithms, Data Structures, Stochastic Processes, Recommender Systems

TECHNICAL SKILLS

Languages: Java, Python, C/C++, SQL, JavaScript, Dart, MATLAB, Stata

Frameworks: TensorFlow, PyTorch, Keras, LangChain, scikit-learn, pandas, NumPy, Apache Spark, Databricks, Flutter

Tools: Cursor, MCP(Model Context Protocol), GCP, Vertex AI, PostgreSQL, Nautilus, MongoDB, LoRA, DPO,

SQLite, Docker, Git, Nautilus, VizTracer, HPC, TPU/V100 GPU, Jupyter, VS Code

EXPERIENCE

AI Researcher

Remote

University of Wisconsin–Madison

June 2026 – Present

- Reduced manual debugging time by 38% by architecting a **LangChain** pipeline routing literature parsing, code generation, and execution across **7** agents into parameterized **PhysiCell** simulations using **NCBI** literature.
- Improved code-generation accuracy by extending a **RAG** retrieval layer over simulation-library Java API docs and reference manuals, and scoping an **MCP server** to standardize tool access across the agent pipeline.

Machine Learning Researcher

San Diego, CA

Halcioğlu Data Science Institute

March 2026 – June 2026

- Designed a novel **bandit-based selection** strategy with **divide-and-conquer decomposition** for **SkyDiscover**, an **LLM-driven** algorithm-discovery framework. Allocated compute more efficiently toward promising solution candidates than existing search strategies (**EoH**, **FunSearch**).
- Evaluated the strategy across **200+** optimization tasks, achieving **31%** fewer **LLM** calls and **24%** faster convergence, orchestrating program generation over a **SQLite** strategy database with **unit tests** and **VizTracer**.

Machine Learning Researcher

San Diego, CA

Human-centered eXtended Intelligence (UCSD Lab)

September 2025 – Present

- Built Reddit-based pipelines to train LLMs toward regulated responses across **CERQ**(cognitive emotion regulation) using **Vertex AI** for preference-pair generation, improving classification by **21** percentage points.
- Fine-tuned **transformer models** on **100K+** samples using **DPO** on **4x A100** Nautilus clusters, with responses preferred in **72%** of human evaluations for emotional regulation across regulated and dysregulated variants.

Machine Learning Intern

New York, NY (Hybrid)

Myna Voice Labs (Research Collaboration with Princeton PhD Scholars)

August 2025 – April 2026

- Achieved **89%** accuracy with strong cross-linguistic generalization by developing Parkinson's disease detection models from speech using **SVMs**, **CNNs**, and **Audio Spectrogram Transformers (AST)**.
- Confirmed model robustness with a **20–28x** separation in confidence interval width between class predictions by evaluating audio perturbations and noise using **Bayesian intervals** across **OpenL3** and **Whisper** feature representations; integrated **LoRA-based continual learning** to retain **95.6%** of prior-task performance.

PROJECTS

Skin Lesion Detection System | *PyTorch, YOLOv8, Docker, Kubernetes*

- Led **9-member** team building an end-to-end skin lesion classification pipeline on **HAM10000**, benchmarking a custom **CNN** against **EfficientNet-B0**, **ResNet18**, and **YOLOv8s**, reaching **81.75%** top-1 accuracy.
- Built **GPU training infrastructure** with **Docker** and **Kubernetes** on **A100** nodes, supporting multi-GPU training, persistent checkpoint storage, and interactive debugging pods.

RAG Pipeline for Internal Query Answering | *LangChain, sentence-transformers, MongoDB, FastAPI*

- Built **RAG** pipeline to replace manual internal doc search, using **LangChain** and **sentence-transformers**.
- Designed data layer with **MongoDB Vector Search** for embeddings and document storage, orchestrated via **FastAPI** and deployed with **Docker**, sustaining **100+** concurrent requests at **p95 latency of 1–2 seconds**.

AI-Integrated Island for Eutrophication Control | *Reinforcement Learning, Python* | [Research Paper](#)

- Developed AI island research prototype using Bellman equation targeting high-phosphate regions; **75%** reduction.
- Implemented value-state matrix for intelligent navigation across grid-based lake environments.